

Diffusion-Autoregression Hybrid Model for Image Generation

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Abstract

We aim to explore a hybrid generative model combining diffusion and autoregressive techniques: run diffusion denoising for a fixed number of steps, then switch to autoregressive refinement on the partially de-noised image. We want to see whether this “diffuse first, AR second” approach can yield better tradeoffs (quality, speed, sample fidelity) than pure diffusion or pure autoregressive models. Our refiner is trained independently as a degradation-to-sharp reconstructor and is applied to diffusion samples at inference time. On CelebA64, qualitative results suggest the AR stage tends to sharpen diffusion outputs and enhance local detail, but can also over-sharpen due to input distribution mismatch.

1. Introduction

Recent progress in generative image modeling has been defined by two primary techniques: diffusion models and autoregressive (AR) models. Diffusion models have produced results that maintain global coherence and are semantically correct by iteratively denoising samples from Gaussian noise [2]. However, these models often involve hundreds or thousands of inference steps, which makes them computationally expensive. This deeply limits usability, especially in compute- or latency-constrained environments.

AR generates images through sequential prediction, where pixels or tokens are conditioned on previous outputs. While this process is faster and results in sharper local detail, AR-generated samples often have less global structure and coherence [6, 11]. This tradeoff between computational cost and image quality motivates our study of a novel hybrid technique. We leverage diffusion to capture the coarse global structure and use an autoregressive model to fill in the fine-grained details.

This differs from prior hybrid generation approaches, which consist of AR image generation followed by diffusion refinement or diffusion interleaved as AR. We observe whether autoregressive modeling can be applied following partial diffusion to sharpen details while preserving

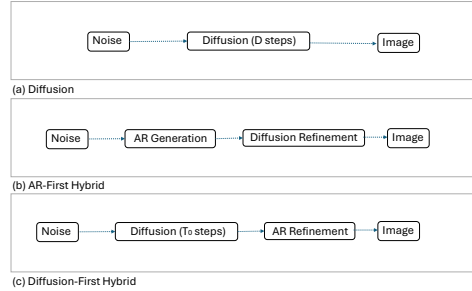


Figure 1. **Architectural comparison of generation strategies.** (a) Pure diffusion iteratively denoises from noise. (b) Prior hybrid designs generate images autoregressively before diffusion refinement. (c) Our two-stage approach switches from diffusion to autoregressive refinement at switch point T_0 .

the global structure produced by diffusion. Our primary research questions are as follows:

- At what noise level (i.e., after how many diffusion steps) should one switch to AR to maximize the quality/time trade-off?
- Does AR finishing sharpen details or correct diffusion artifacts?

To study this, we train a diffusion model and a lightweight AR decoder independently on the CelebA dataset and evaluate the image quality after different numbers of time steps. The AR model works on partially denoised images as input to refine their structure. We evaluate results by analyzing the visual quality of samples and artifact behavior.

2. Related Work

2.1. Diffusion Models

Denoising Diffusion Probabilistic Models (DDPM) demonstrated that diffusion through iterative denoising can generate high-quality images, achieving strong results on CIFAR-10 [2]. More recently, Denoising Diffusion Implicit Models (DDIMs) have been capable of high-quality image generation with fewer steps through a non-Markovian sampling

process [8]. While these approaches improve efficiency, too significant of a reduction in steps can reduce sample diversity or image quality.

More recently, research has been performed in hosting diffusion models in compressed latent spaces [7]. While this reduces the overall computational cost, it still requires iterative denoising. Consequentially, diffusion models still face a tradeoff between sampling speed and visual quality. Central to latent diffusion and other latent-space generative methods are variational autoencoders (VAEs), which learn a continuous latent variable model for images [3]. Although VAEs create an efficient latent space, they often provide poor image quality in terms of blur. Consistency models learn mappings between noisy and clean images, which involves no iterative denoising [9]. Therefore, inference time is significantly reduced compared to standard diffusion. Advancements such as consistency models still do not attempt to correct artifacts near the end of generation as we do with diffusion denoising.

2.2. Autoregressive Models

Autoregressive (AR) image models generate outputs sequentially, either pixel-by-pixel or token-by-token. Pixel-CNN was an early example that generated every pixel in raster order, but it suffered from slow sampling and poor scalability [11]. More recent autoregressive methods instead use discrete tokens; for example, DALL-E uses a 32×32 grid of tokens to represent a 256×256 image [6]. This speeds up sampling considerably, but compression into discrete codes introduces errors and can produce artifacts or blurriness [10]. Recent work in AR generation without vector quantization has improved quality while utilizing sequential generation [4]. Autoregression is faster than diffusion when using discrete image patches, but often worse in terms of fine detail and global coherence.

2.3. Hybrid AR-Diffusion Models

2.3.1. AR-First Hybrids

Hybrid approaches try to combine the strengths of both. HART [10] first uses an autoregressive transformer to generate discrete tokens for a coarse image, and then applies a small diffusion model to refine details. This made generation five to nine times faster while still improving fidelity compared to pure AR. Another method replaces discrete tokenization with continuous-valued vectors and trains the AR model with a Diffusion Loss function [4]. This prevents quantization errors that can result in blur in discrete-token AR models, while still maintaining efficiency benefits of token-level generation.

2.3.2. Interleaved Models

Another line of hybrids reframes diffusion as an autoregressive process. D-AR tokenizes the denoising steps and gen-

erates them sequentially with a transformer [1]. This design mimics diffusion in one pass and reaches image quality close to standard diffusion models. Methods such as these unify diffusion and autoregression into a single framework as opposed to using independently trained stages.

In contrast to autoregressive generation followed by diffusion or reinterpretation of diffusion as AR with tokenization, we study a two-stage pipeline. First, our work leverages diffusion to construct the global image structure. Then, we apply an AR to refine the high-frequency details and correct artifacts in partially denoised images. Figure 1 shows the novelty of our work as compared to previous architectures. Our results suggest that autoregression can improve the sharpness of diffusion images, which highlights the complementary role of these paradigms.

3. Approach

Our system is trained as two separate models: (1) a baseline diffusion generator trained in the standard DDPM setting, and (2) an autoregressive (AR) refiner trained to reconstruct sharp images given a degraded conditioning image. At inference time, we replace the degradation conditioning with a partially denoised diffusion sample.

3.1. Dataset

All experiments use face images from the CelebA dataset. We utilize the Hugging Face `nielsr/CelebA-faces` repository, resized to 64×64 resolution. Detailed preprocessing, normalization steps, and split configurations are provided in Appendix B.

3.2. Diffusion Image Generator

We train a standard conditional DDPM [2] with a U-Net backbone (64×64 resolution, ≈ 50 M parameters). The model optimizes the standard mean-squared error objective over 1,000 diffusion timesteps using a linear beta schedule. Detailed architectural specifications and training objectives are provided in Appendix A.

3.3. Autoregressive Refinement Model

The autoregressive refiner is trained independently from the diffusion backbone. It functions as a sequence-to-sequence model that maps a degraded input to a high fidelity output.

Data Degradation Strategy. During training, for each ground-truth image x , we generate a conditioning input $\tilde{x}$ by applying a stochastic degradation pipeline. As implemented in our dataset logic, we randomly sample from a set of degradations including Gaussian blur, bicubic downsampling (to resolutions as low as 32×32), and JPEG compression. This prevents the model from learning a simple identity mapping and forces it to learn high-frequency details.

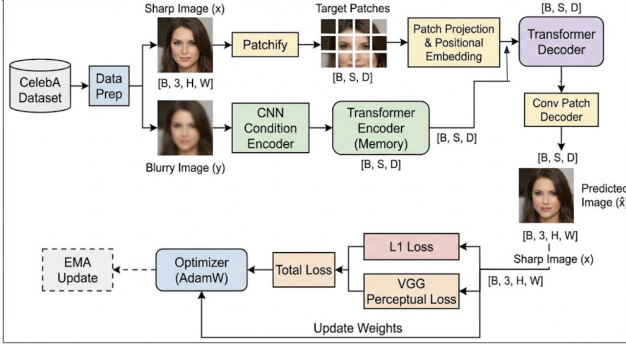


Figure 2. Autoregressive refinement model trained on degraded→sharp reconstruction (degradation serves as a proxy for partially denoised diffusion outputs).

Conditioning Pathway. The degraded image $\tilde{x}$ is processed by a CNN encoder to extract a global context sequence. The encoder consists of four convolutional layers using Group Normalization and ReLU activations. The first two layers use stride 2 to downsample the resolution from 64×64 to 16×16 , while the subsequent two layers maintain resolution to refine features. The resulting 16×16 feature map is flattened and processed by a 2-layer Transformer encoder to aggregate global spatial context.

Transformer Backbone. We treat the target image x as a sequence of flattened 4×4 patches. These patches are projected via a linear layer and normalized with LayerNorm. We add learnable positional embeddings to preserve spatial information. The core backbone is a Transformer Decoder (6 layers, 4 heads, 256 embedding dimension) which predicts the sequence autoregressively. The decoder utilizes causal masking to ensure the prediction of patch t depends only on patches $0 \dots t - 1$ and the global context from the conditioning pathway.

Spatially-Aware Decoder Head. Standard vision transformers regress pixels using a simple linear projection. To mitigate blocking artifacts, we employ a convolutional head. The transformer output vectors are first expanded via a 2-layer MLP, reshaped into spatial feature maps, and then upsampled using Transposed Convolutions. A final Tanh activation ensures the output lies in the range $[-1, 1]$.

Training Objective. The model is optimized using a compound loss function:

$$\mathcal{L}_{\text{total}} = |x - \hat{x}|_1 + \lambda_p \mathcal{L}_{\text{VGG}}(x, \hat{x}) \quad (1)$$

We use an L1 pixel reconstruction loss combined with a VGG-16 perceptual loss ($\lambda_p = 0.15$). The perceptual loss computes the Mean Squared Error between feature maps extracted from the first 16 layers (up to relu3_3) of a pre-trained VGG network.

Optimization & Scheduled Sampling. We train using AdamW (lr = 3×10^{-4} , weight decay 10^{-3}) with a co-

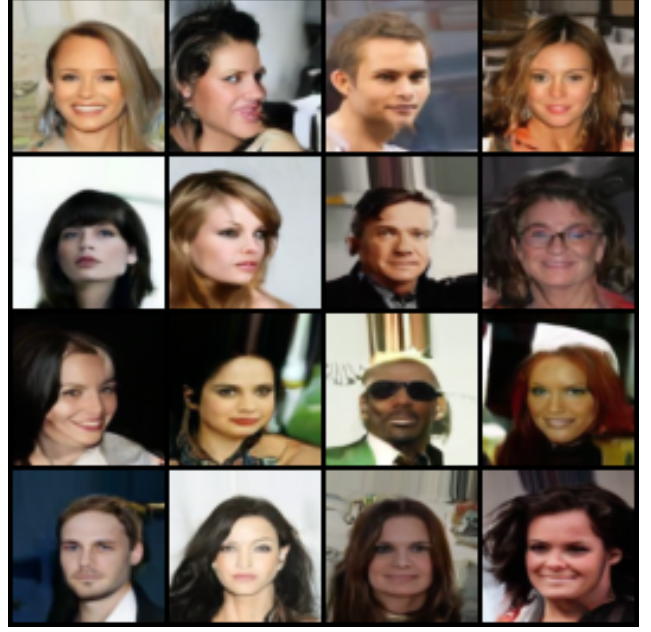


Figure 3. Diffusion image generation results after 42,000 training steps (28 epochs).

sine annealing schedule following 1000 warmup steps. To bridge the train-test gap caused by exposure bias, we implement scheduled sampling. From epoch 5 to 15, we linearly increase the probability (from 0% to 30%) of feeding the model’s own generated patches as context for the next step, rather than the ground truth patches (teacher forcing).

3.4. Compute and Runtime Considerations

Diffusion sampling requires T U-Net forward passes (one per denoising step). AR refinement requires autoregressive decoding over $S = 256$ patches; in our setup, this decoding is fully sequential, requiring S transformer-decoder forward passes. Our AR model is only 14M parameters, (3.5x smaller than diffusion model) and allow for fewer diffusion steps while maintaining perceived sharpness.

4. Results

4.1. Diffusion Baseline

Figure 3 shows samples from our baseline diffusion model after training for 42,000 steps (28 epochs) using the DDPM objective.

4.2. Autoregressive Decoder (Standalone)

We validate the AR decoder in isolation. The model is conditioned on a degraded (blurred/noisy) input image and trained to reconstruct the corresponding clean image. Figure 4 visualizes the behavior.



Figure 4. AR decoder training result (standalone). Top: blurred/noisy input; Middle: clean target; Bottom: model reconstruction.

4.3. Hybrid: Diffusion Followed by AR Refinement

We evaluate the two-stage pipeline by running diffusion for a fixed number of denoising steps T , then applying the AR refiner to the partially denoised output. Figure 5 shows qualitative comparisons at 1000 steps. Results at various denoising steps are provided in Appendix C.

Across settings, the AR stage primarily sharpens the diffusion outputs, which is consistent with how it is trained (degradation-to-sharp reconstruction). In some cases the refinement appears over-sharpened; we attribute this to mismatch between the degradation distribution used during AR training and the residual diffusion artifacts present at finite T .

4.3.1. Quantitative observations.

We compute pixel-level statistics to characterize refinement behavior. Across batches, the average L1 difference between diffusion outputs and AR-refined images is approximately 0.08–0.12, indicating moderate but noticeable changes. Refined images typically exhibit slightly higher pixel-value standard deviation (e.g., 0.28 vs. 0.26), consistent with sharpening and contrast enhancement. These metrics confirm that the AR stage applies substantive modifications rather than identity-like mappings.

4.3.2. Simple baseline comparison (discussion).

A key question is whether the AR stage provides value beyond classical post-processing (e.g., an unsharp mask). While we do not include a full baseline sweep, future work should compare against simple filters on the same diffusion outputs to isolate when refinement is truly semantic/detail-correcting rather than purely sharpening.

4.3.3. Failure modes and limitations.

Observed/expected failure cases include: (i) over-sharpening and halos, (ii) texture distortion where high-frequency detail is amplified unnaturally, and (iii) domain limitation since all training/evaluation is on CelebA faces at 64×64 .



Figure 5. Hybrid results. Top: diffusion output ($T = 1000$ steps). Bottom: AR refinement.

5. Conclusion

Our experiments suggest that applying an autoregressive refiner after diffusion denoising can sharpen diffusion outputs and enhance local detail on CelebA at 64×64 . However, the current refiner is trained on a degradation proxy rather than true diffusion residuals, and our results are qualitative; we therefore treat the method as a promising direction rather than a demonstrated improvement in speed–quality trade-offs.

Future work includes:

- jointly training diffusion and AR end-to-end,
- training the AR refiner directly on partially denoised diffusion samples to better match inference-time inputs,
- scaling the AR refiner capacity,
- exploring token-based pipelines (e.g., VQ-VAE) for efficiency,
- quantitative comparisons (e.g., FID vs. diffusion timesteps T) at matched compute, and
- adding non-learned baselines (e.g., unsharp mask) to test whether gains persist beyond basic sharpening.

References

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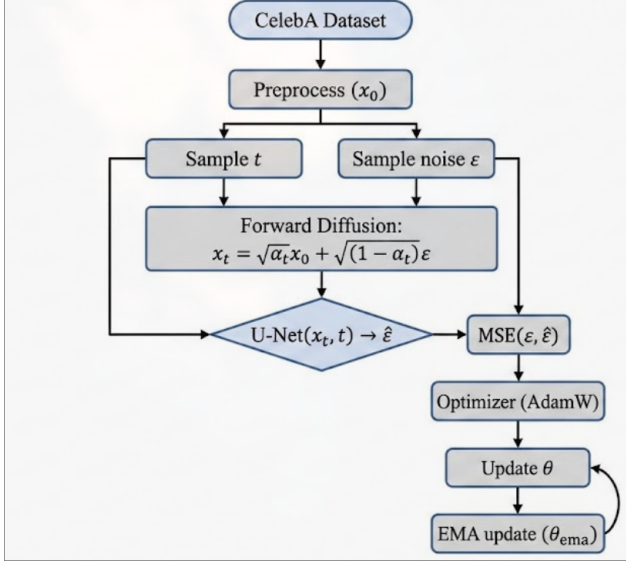


Figure 6. Diffusion image model training and sampling pipeline.

A. Diffusion Model Implementation

We train a basic image diffusion model following the DDPM formulation [2]. The model learns to predict noise under a mean-squared error objective. The standard forward diffusion process is:

$$x_t = \sqrt{\alpha_t}x_0 + \sqrt{(1 - \alpha_t)}\epsilon. \quad (2)$$

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{x_0, t, \epsilon} [\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2]. \quad (3)$$

Model and schedule. Our diffusion model uses a U-Net backbone at 64×64 with one residual block per resolution stage and channel widths (128, 256, 256, 512). We include self-attention in the deeper encoder/decoder stages (lower spatial resolutions). The model contains approximately 50M trainable parameters. We train with 1000 diffusion timesteps and a linear beta schedule.

Optimization and training. We train with AdamW [5] ($\text{lr} = 10^{-4}$, weight decay 0.01, betas (0.9, 0.999), $\epsilon = 10^{-8}$) and a warmup+cosine learning-rate schedule (500 warmup steps) with gradient clipping (1.0). Training is run for 28 epochs with batch size 128 and EMA (decay 0.9999). Mixed precision (FP16) is enabled.

Accelerated inference with DDIM. At inference time, we optionally replace the DDPM sampler with DDIM [8], reducing from 1000 to as few as 4 denoising steps ($\sim 250\times$ speedup) with minimal quality degradation. This enables rapid iteration when evaluating the hybrid pipeline.

B. Dataset and Preprocessing Details

We perform the following preprocessing steps for the CelebA dataset:

- **Diffusion training data:** We train on the Hugging Face dataset `nielsr/CelebA-faces`, resized to 64×64 , with random horizontal flips for data augmentation. Images are converted to tensors and normalized to $[-1, 1]$ using mean (0.5, 0.5, 0.5) and standard deviation (0.5, 0.5, 0.5).
- **AR refiner training data:** We train the refiner on CelebA face images resized to 64×64 . Images are loaded from a local image folder, converted to tensors in range $[0, 1]$, and then normalized to $[-1, 1]$ via the transformation $(x - 0.5) \cdot 2$ for both the degraded/conditioning input and the target.
- **Train/val split:** The AR dataset is split 90/10 using a fixed random seed (42).

C. Results at various denoising steps



Figure 7. Hybrid results at multiple diffusion budgets. Top: diffusion output ($T = 600$ steps). Bottom: AR refinement. Top: diffusion output ($T = 800$ steps). Bottom: AR refinement.

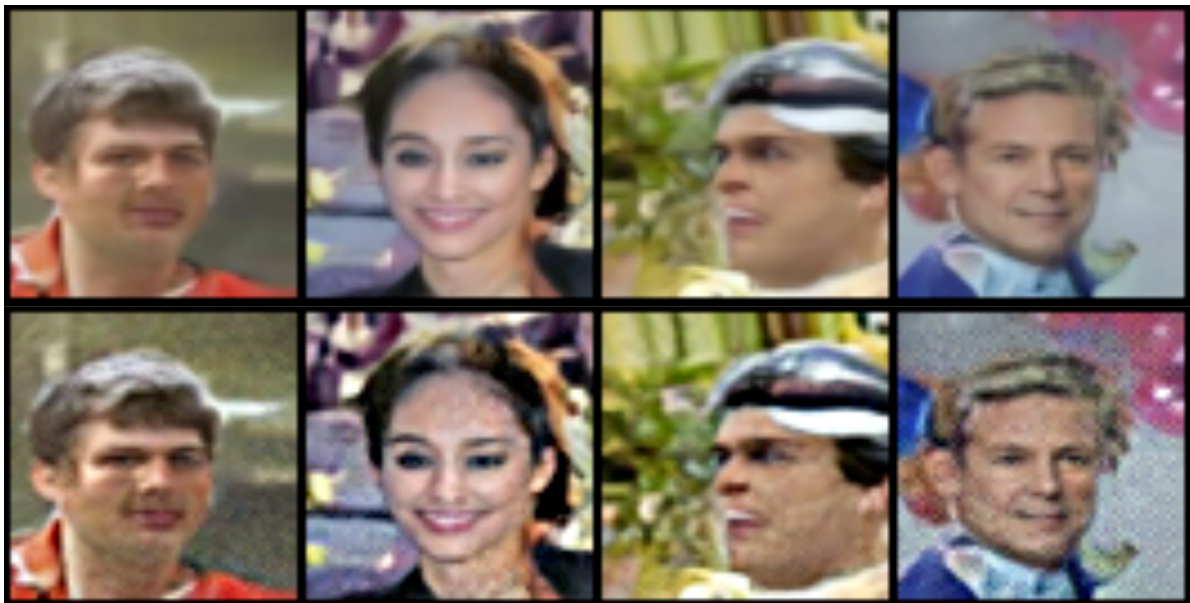


Figure 8. Hybrid results at multiple diffusion budgets. Top: diffusion output ($T = 800$ steps). Bottom: AR refinement.